

**STELLENBOSCH UNIVERSITY
DEPARTMENT OF ECONOMICS**

ECONOMICS 114

MAIN ASSESSMENT A1: 5 APRIL 2022

Time: 2.5 hours

Marks: 80

EXAMINERS:

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INTERNAL MODERATOR: Dr E Moses

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS NEXT** TO THE NUMBER OF THE STATEMENT.

KEY:

AP = Average product

BL = Budget line

GDP = Gross domestic product

IC = indifference curve

K = capital

FF = Feasible frontier

MP = Marginal product

MRS = Marginal rate of substitution

MRT = Marginal rate of transformation

P = Price

PF = Production frontier

Q = Quantity

TP = Total product

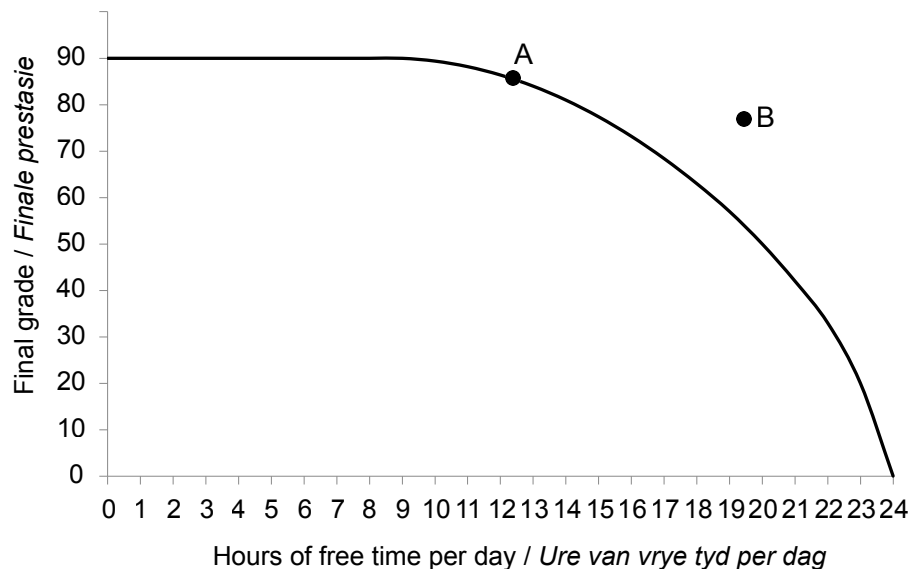
w = Wage

Y = Income

SECTION A
TRUE/FALSE STATEMENTS

[6 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. The growth of firms and the expansion of markets during the Industrial Revolution allowed for large-scale specialisation.
a) True
b) False
2. In the Malthusian model, the production function is characterised by diminishing average product of labour.
a) True
b) False
3. Jim receives a wage of R20 per hour, and Dwight receives a wage of R15 per hour. To maximise utility, Jim works 35 hours per week and Dwight works 42 hours per week. The opportunity cost of an additional hour of free time is higher for Dwight than it is for Jim.
a) True
b) False
4. The figure below indicates the feasible set of a student's final grade and hours of free time per day. Based on this information, we can say that the student's choice between A or B depends on their preferences.



- a) True
b) False
5. A strategic interaction in which two self-interested individuals act independently always leads to a socially sub-optimal outcome.
a) True
b) False

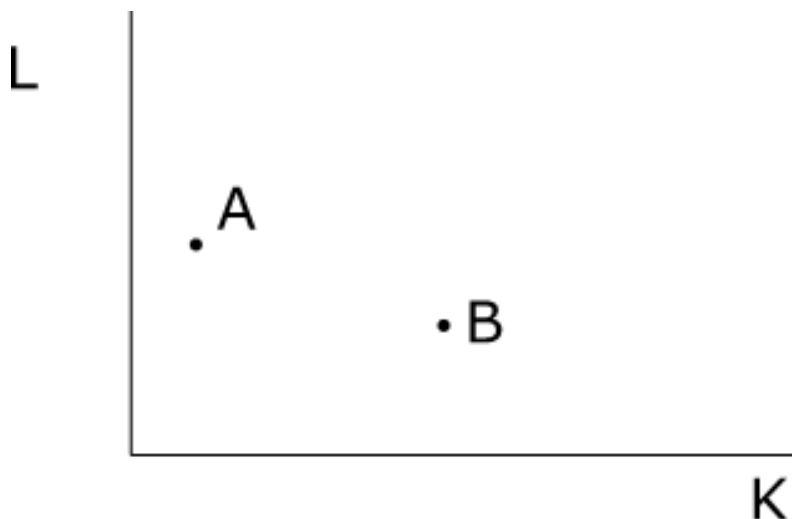
6. Procedural fairness implies substantive fairness.
- a) True
 - b) False

SECTION B

MULTIPLE CHOICE QUESTIONS

[24 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

7. The kitchen manager at an Italian restaurant is deciding what tasks to assign to his two cooks, John and David. John can make 25 pizzas or 40 servings of pasta per hour and David can make 20 pizzas or 30 servings of pasta. Which of the following should be the manager's choice?
- a) Fire David because he is not as productive as John. John will do both jobs.
 - b) John will make pizza because he has a comparative advantage in making pizza.
 - c) David will make pizza because he has a comparative advantage in making pizza.
 - d) John and David both will spend half their time making pizza and half their time making pasta because both have a comparative advantage in making pizza.
8. Which of the following economic institutions play an important role in capitalism as an economic system?
- a) Markets and Government
 - b) Government and Private property
 - c) Families
 - d) Private property and Markets
9. In the graph below, K is the amount of capital required in production and L the amount of labour required in production, and A and B represent different technologies for producing 100 cucumbers.



B is currently the cheapest technology. However, A could become the cheapest technology if

- a) the price of capital and labour both decrease by the same percentage.
 - b) the price of capital and labour both increase, with a greater percentage increase for the price of labour.
 - c) the price of capital and labour both decrease, with a greater percentage decrease for the price of capital.
 - d) the price of capital increases while the price of labour remains constant.
10. Zuri and Nala spend their time producing tomato sauce and mustard. The following table indicates the maximum amount of each product that can be produced if they spend an hour on producing only that good.

	Tomato sauce (bottles)	Mustard (bottles)
Zuri	6	3
Nala	12	4

- Which person has the comparative advantage in the production of tomato sauce?
- a) Zuri
 - b) Nala
 - c) Both Zuri and Nala
 - d) Neither of them
11. Ratul consumes a combination of instant noodles (an inferior good) and Oreos (a normal good). Ratul has an income of R100 per day to spend on these two goods. Due to his diligence at work, his employer decides to give him a raise of R50 per day. Assume the prices of instant noodles and Oreos remain the same. Which of the following must be TRUE?
- a) Ratul's budget constraint remains the same after the income increase as it was before.
 - b) Ratul's consumption of instant noodles will increase.
 - c) Ratul will no longer consume any instant noodles.
 - d) Ratul's budget constraint will shift parallel outwards.
12. Vuyokazi consumes only chocolate and magazines. Which of the following will lead to a change in the shape of her indifference curves?
- a) A change in her preferences for either chocolates or magazines, or both.
 - b) A change in the price of either chocolates or magazines, or both.
 - c) A change in her income.
 - d) Both (b) and (c) are true.
13. Assume on Ragnarök that free time is an inferior good. Suppose that Loki's wage rate falls. Which one of the following must be TRUE?
- a) Loki will work more because the income effect on free time dominates the substitution effect on free time.
 - b) Loki will work more because the substitution effect on free time dominates the income effect on free time.

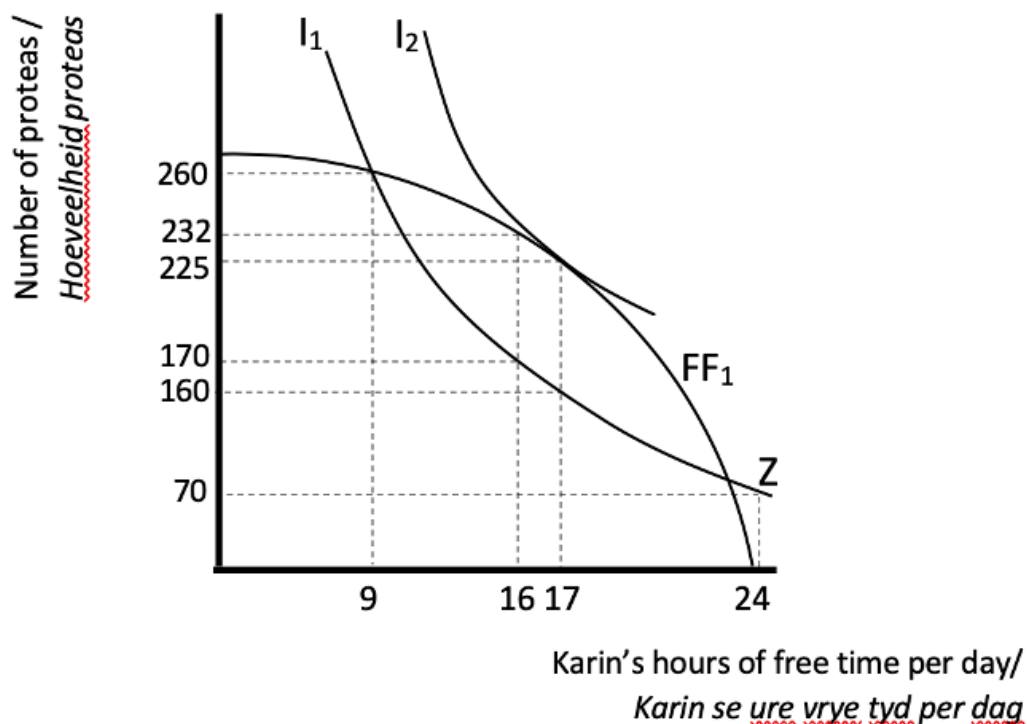
- c) Loki will not change his hours of work because the substitution and income effects on leisure exactly offset each other.
- d) Loki will work less, because both the income and substitution effects cause him to choose more free time.
14. Luke's final performance in Accounting is determined by a strictly concave production function in which hours of study is the input. Luke will achieve a final mark of 0% if he spends no hours studying. Which one of the following statements regarding Luke's average product (AP) and marginal product (MP) of time spent studying is TRUE?
- a) MP increases with each additional hour spent studying.
- b) MP remains constant no matter how many hours are spent studying.
- c) Luke's AP of studying always exceeds his MP of studying.
- d) Luke's MP of studying always exceeds his AP of studying.
15. Which one of the following defines a dominant strategy best?
- a) A strategy that gives a player the highest possible payoff, given the strategies that the other player selects.
- b) A strategy that results in the player having no incentive to change their strategy given the strategy of the other player.
- c) A strategy that gives a player the highest possible payoff, irrespective of the strategies that the other player selects.
- d) All of the above are correct.
16. Ben and John are working together on a class presentation. They have a choice between putting in high effort, or putting in low effort and free riding on the other player's effort. The more effort they invest in the presentation, the higher the mark they will receive. The payoff matrix below represents this strategic interaction:

		John	
		Low effort	High effort
Ben	Low effort	5, 5	8, 4
	High effort	4, 8	7, 7

- Which one of the following statements are CORRECT?
- a) John's best response if Ben plays 'low effort' is 'high effort'.
- b) John's best response if Ben plays 'high effort' is 'high effort'.
- c) John's best response if Ben plays 'low effort' is 'low effort'.
- d) Ben's best response if John plays 'high effort' is 'high effort'.
17. Which one of the following statements is TRUE?
- a) A factory owner will have low bargaining power in wage negotiations when they employ many people.

- b) Suppose there are two possible allocations, A and B, and there are only two people in the world. If person 1 prefers A over B and person 2 prefers B over A, then both allocations are Pareto inefficient.
- c) When two people are given a resource and the resource is divided amongst both equally, the allocation is always Pareto efficient.
- d) In an economic exchange, bargaining power is the extent of a person's advantage in securing a larger share of economic rents.

18. Karin grows King proteas on Pam's land but has to pay Pam a portion of her crop for the right to do so. Maximum possible production is represented by the combined feasible frontier (FF_1) and Karin's reservation indifference curve (I_1). Karin's preferences are quasilinear. Which one of the following statements is INCORRECT?



- a) Pam will maximise her surplus when she asks for rent of 65 proteas and Karin chooses to work 7 hours a day.
- b) Karin would never choose to work more than 15 hours a day.
- c) The number of proteas that Karin would have to sacrifice exceeds the number of proteas that Karin is willing to sacrifice when she increases her hours of free time from 16 to 17.
- d) It would be Pareto efficient for Karin to work 7 hours a day and to pay Pam 50 proteas.

SECTION C
LONG QUESTIONS
[50 marks]

QUESTION 1

- 1.1. The table below includes the nominal GDP values for South Africa for the period 2005 to 2010. What was the nominal GDP growth rate for the period 2008 to 2010? [2]

South Africa Annual gross domestic product at nominal prices (R million)						
Year	2005	2006	2007	2008	2009	2010
GDP	1 639 254	1 839 401	2 109 501	2 369 063	2 507 677	2 748 008

- 1.2. Provide a critique of the use of GDP as a measure of the standard of living (wellbeing) over time. [2]

QUESTION 2

A firm makes use of combinations of labour (L) and capital (K) to produce its output. Suppose that the wage rate (w) — the price of labour — is R10 and that the price of a unit of capital is R20. A firm has the option to choose between three technologies (A, B and C) for producing a particular quantity of its output. The inputs required for each of these technologies are:

- A: 50L and 2K
B: 20L and 12K
C: 5L and 16K

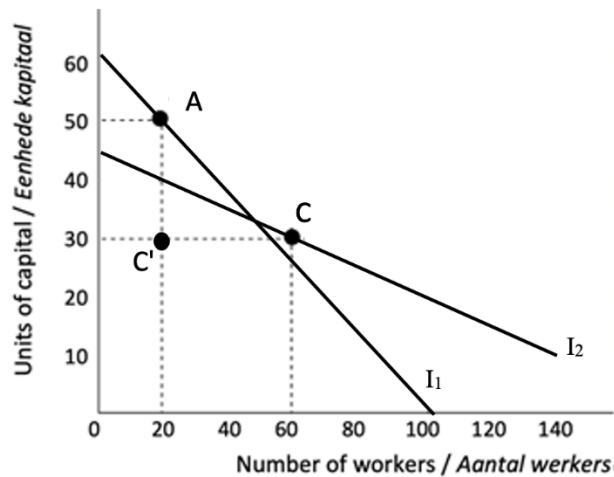
- 2.1 If capital is plotted on the vertical (y-) axis, derive the equation of the isocost line that corresponds to a total cost of R100. [1]
2.2 Write down the slope of your isocost line from 3.1. What does the magnitude (absolute value) of the slope represent? [2]
2.3 Which of technologies A, B or C will the firm choose to use to produce their output? Show your calculations. [2]

QUESTION 3

- 3.1 The Malthusian model explains why living standards remained at or near subsistence levels for centuries before the Industrial Revolution. Explain why this happened (no graphs required). [3]

Consider the figure below which indicates three different technologies, A, C and C', that could be adopted by Chinese and Polish factories to produce 500 wooden picture frames per day. Labour is relatively more expensive in Poland than it is in China such that isocost line I_1 is drawn for the relative costs of labour and capital in Poland, whilst isocost line I_2 is drawn for

the relative costs of labour and capital in China. Assume that technology C' has not yet been developed.



3.2 Which of the technologies, A or C, is China more likely to adopt? Provide a reason for your answer. [2]

Technological innovation in Poland means that technology C' is now feasible for factories to use to produce picture frames. Suppose that the wage paid to labour in Poland is 10 złoty and the cost per unit of capital is 120 złoty.

3.3 How much innovation rent do early adopters of this technology in Poland stand to make? Assume that the price of a picture frame remains the same. [1]

QUESTION 4

A consumer has a total budget of R570 to spend on bananas (B) and sunflowers (S). A banana costs R5 and a sunflower costs R10. Both are considered to be normal goods by the consumer.

4.1 What is the consumer's marginal rate of substitution at their optimal point if the quantity of bananas is plotted on the vertical axis? [2]

4.2 How many bananas will the consumer be willing to trade for an additional sunflower at the optimal consumption point? [1]

The price of bananas now increases to R7.50 each.

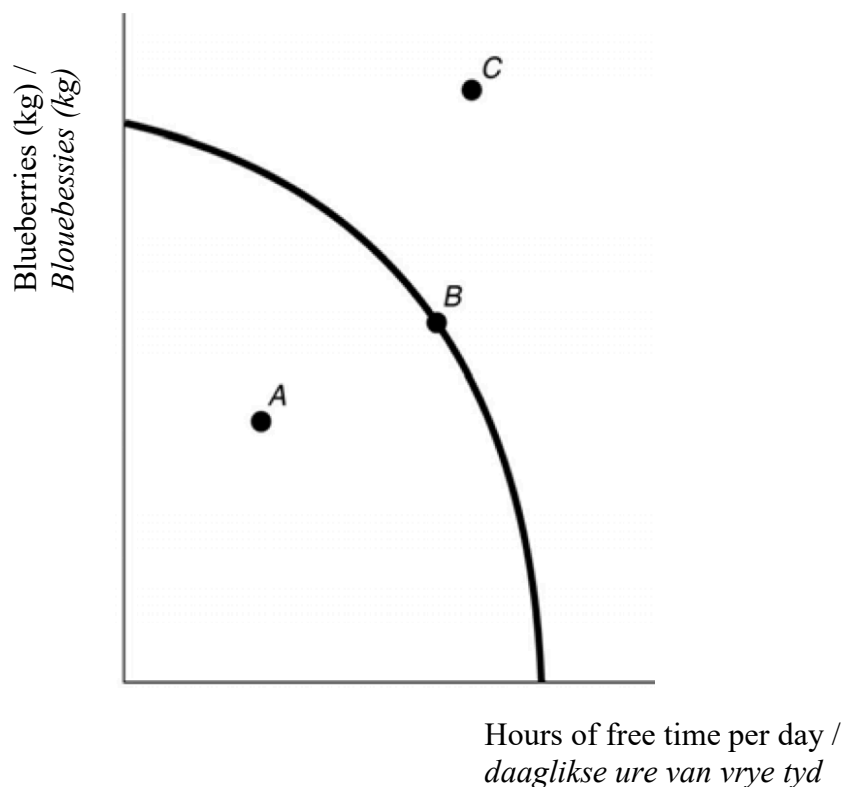
4.3 How has the consumer's real income in terms of sunflowers been affected? Show your calculations. [2]

4.4 What will the consumer's MRS be at their new optimal consumption point? [1]

4.5 Do you expect that the consumer will purchase more or fewer sunflowers? Explain. [3]

QUESTION 5

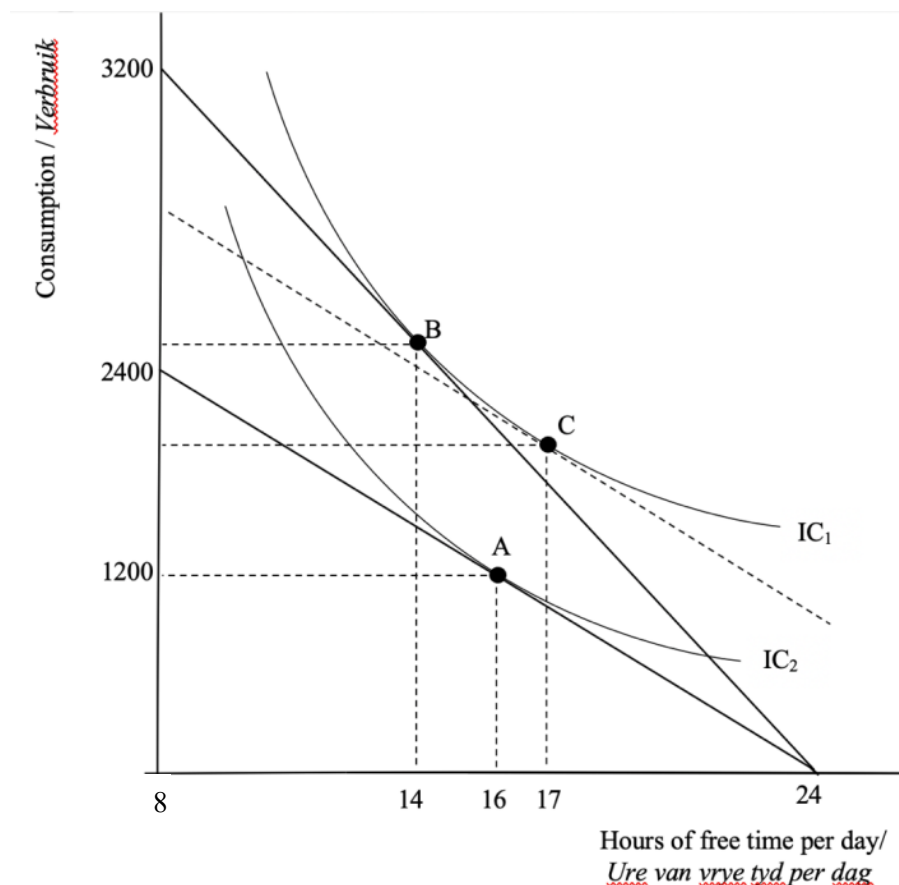
The graph below indicates the feasible frontier of Sally, a subsistence farmer who grows and harvests blueberries.



- 5.1 Which of the combinations **A**, **B** and **C** are feasible? [1]
- 5.2 What is the opportunity cost of an extra hour of free time at point **A**? [1]
- 5.3 Would Sally prefer to be at point **B** or point **C**? Explain. [2]
- 5.4 How would the introduction of a serious insect pest affect Sally's feasible set? [2]

QUESTION 6

You are working as an intern during the summer holidays and currently optimise your labour-leisure decision at point **A** (note: the graph is not drawn to scale). You only have labour income with which to spend on consumption.



6.1 What is your current hourly wage rate? [1]

Your employer is impressed with the quality of your work and decides to increase your hourly wage to R200. This takes you to a new optimal point **B**.

6.2 What are the income and substitution effects on your free time? Provide economic reasoning for the directions of these effects. [4]

QUESTION 7

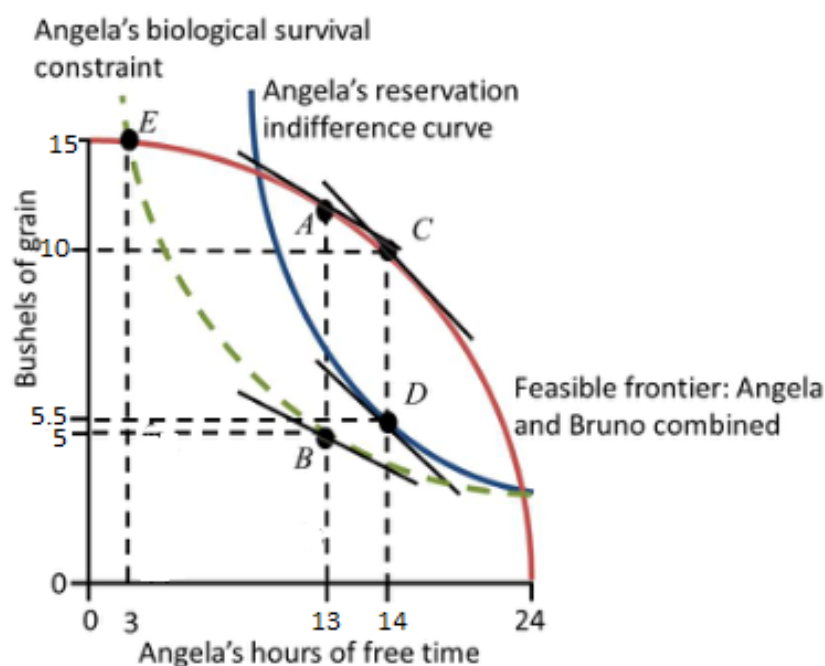
Mary and Joan want to meet up to study together for an upcoming test. They have a choice between meeting in the library or the local coffee shop. Mary prefers meeting in the library, whilst Joan prefers meeting in the local coffee shop. However, both prefer to go to the same place so that they can study for the test together.

		Joan	
		Library	Coffee shop
Mary	Library	5, 3	2, 2
	Coffee shop	0, 0	3, 5

- 7.1 What is the Nash equilibrium in this game? [2]
 7.2 Determine whether Mary and Joan have dominant strategies. Explain why such a strategy does exist or does not exist. [2]
 7.3 Does this game have a dominant strategy equilibrium? Explain why or why not. [1]

QUESTION 8

Bruno is a landowner and Angela is a farmer who pays a share of her grain output to Bruno for the use of the land. The figure shows Angela and Bruno's feasible frontier, Angela's biological survival constraint and her reservation indifference curve. Angela has quasi-linear indifference curves.



- 8.1 Is the following statement true or false? Explain your answer.

“For any given number of hours of free time, the marginal rate of substitution on the reservation indifference curve is smaller than that on the biological survival constraint.” [1]

- 8.2 Assume Angela has some bargaining power, but still has to pay Bruno in grain for the use of his land. How many hours will Angela work and how many bushels of grain will she give to Bruno? [2]
 8.3 Discuss how the equilibrium allocations of grain are reached in the case of a) coercion and b) bargaining power. Compare the size of the surplus between the case of coercion and where there is bargaining power. [3]

QUESTION 9

9.1 Consider an economy that contains a fixed amount of only one good, which everyone likes and all of which is allocated to someone. Explain why every allocation is Pareto efficient.

[2]

9.2 An economy contains two people and two goods, bread and apples. Person 1 likes bread and dislikes apples (the more apples she has, the worse off she is), and person 2 likes apples and dislikes bread. There are 100 loaves of bread and 100 apples available. There is only one allocation in this economy that is Pareto efficient. What is it?

[2]